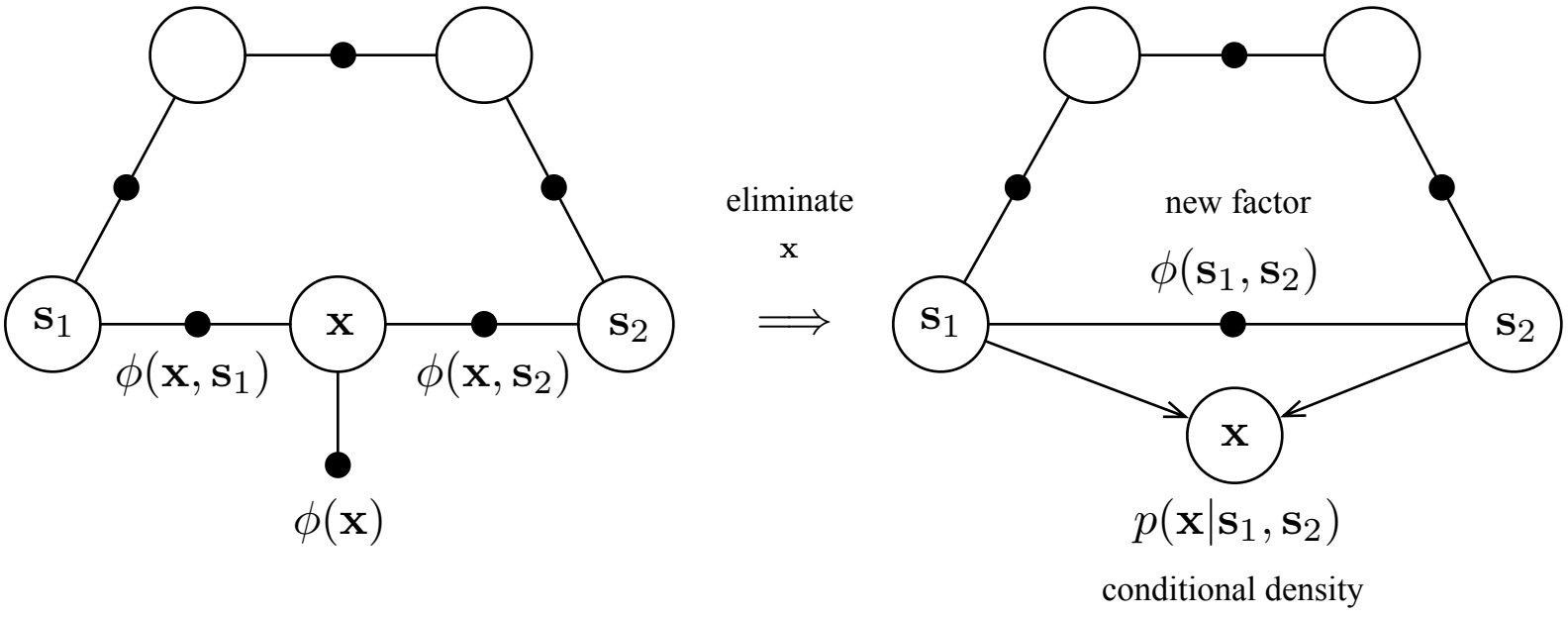
$\mathbf{s}_1$
$\mathbf{x}$
$\mathbf{s}_2$
$\phi(\mathbf{x}, \mathbf{s}_1)$
$\phi(\mathbf{x}, \mathbf{s}_2)$
$\phi(\mathbf{x})$
eliminate
$\mathbf{x}$
$\Longrightarrow$
new factor
$\phi(\mathbf{s}_1, \mathbf{s}_2)$
$\mathbf{s}_1$
$\mathbf{s}_2$
$\mathbf{x}$
$p(\mathbf{x}|\mathbf{s}_1, \mathbf{s}_2)$
conditional density